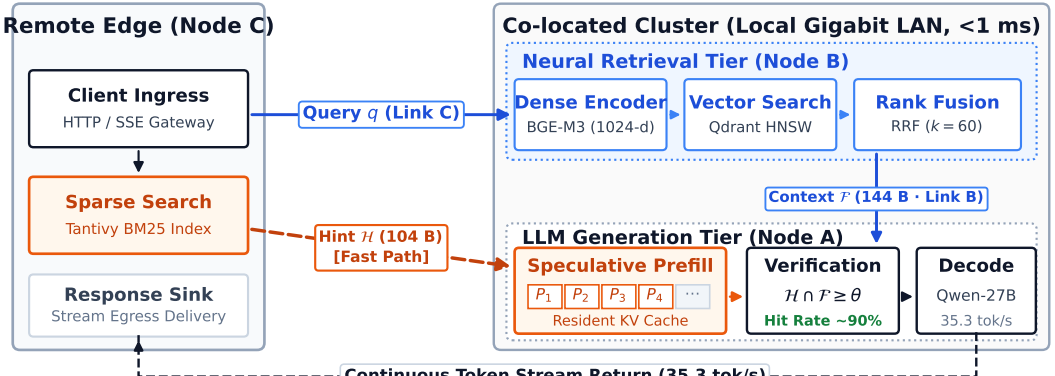
Remote Edge (Node C)
Client Ingress
HTTP / SSE Gateway
Sparse Search
Tantivy BM25 Index
Response Sink
Stream Egress Delivery
Query $q$ (Link C)
Hint $\mathcal{H}$ (104 B)
[Fast Path]
Co-located Cluster (Local Gigabit LAN, <1 ms)
Neural Retrieval Tier (Node B)
Dense Encoder
BGE-M3 (1024-d)
Vector Search
Qdrant HNSW
Rank Fusion
RRF ($k = 60$)
Context $\mathcal{F}$ (144 B · Link B)
LLM Generation Tier (Node A)
Speculative Prefill
$P_1$ $P_2$ $P_3$ $P_4$ ⋯
Resident KV Cache
Verification
$\mathcal{H} \cap \mathcal{F} \geq \theta$
Hit Rate ~90%
Decode
Qwen-27B
35.3 tok/s
Continuous Token Stream Return (35.3 tok/s)